

Creation Date 22-Sep-2009

Revision Date 22-Sep-2009

Revision Number 1

1. IDENTIFICATION OF THE SUBSTANCE/PREPARATION AND OF THE COMPANY/UNDERTAKING

Product Name SAF FIXATIVE
Synonyms No information available.
Recommended Use Laboratory chemicals

Company Remel
12076 Santa Fe Drive
Lenexa, KS 66215 United States
Telephone: 1-800-255-6730
Fax: 1-800-621-8251

Emergency Telephone Number INFOTRAC 1-352-323-3500 (International)
1-800-535-5053 (North America)

2. HAZARDS IDENTIFICATION

Classification according to European directive on classification of hazardous preparations 1999/45/EC.

Xn



R -phrase(s)
none
R40 - Limited evidence of a carcinogenic effect
R43 - May cause sensitization by skin contact

3. COMPOSITION/INFORMATION ON INGREDIENTS

Haz/Non-haz

Component	Weight %	EC No.	Classification
Formaldehyde 50-00-0	1.5	-	Carc.Cat.3;R40 T;R23/24/25 C;R34 R43
Methyl alcohol 67-56-1	0.5	EEC No. 200-659-6	F;R11 T;R23/24/25-39/23/24/25
Acetic acid 64-19-7	2	EEC No. 200-580-7	R10 C;R35
Sodium acetate 127-09-3	1.5	EEC No. 204-823-8	-

SAF FIXATIVE

Revision Date 22-Sep-2009

For the full text of the R phrases mentioned in this Section, see Section 16

4. FIRST AID MEASURES

General Advice	Show this safety data sheet to the doctor in attendance. Repeated contact may cause allergic reactions in very susceptible persons.
Eye Contact	Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Obtain medical attention. May produce an allergic reaction. Rinse thoroughly with plenty of water for at least 15 minutes and consult a physician.
Skin Contact	Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur. Eczema injuries of an allergic nature. Allergic symptoms may develop within 12 hours after exposure.
Ingestion	Clean mouth with water. Obtain medical attention. May produce an allergic reaction. Do not induce vomiting. Drink plenty of water. Never give anything by mouth to an unconscious person. If symptoms persist, call a physician.
Inhalation	Move to fresh air. If breathing is difficult, give oxygen. Get medical attention immediately if symptoms occur. May cause allergic respiratory reaction. If breathing is irregular or stopped, administer artificial respiration. Do not use mouth-to-mouth resuscitation if victim ingested or inhaled the substance; induce artificial respiration with a respiratory medical device.
Notes to Physician	Treat symptomatically.
Protection of First-aiders	Avoid direct contact with skin. Use barrier to give mouth-to-mouth resuscitation. Ensure that medical personnel are aware of the material(s) involved, and take precautions to protect themselves.

5. FIRE-FIGHTING MEASURES

Suitable Extinguishing Media

Use water spray, alcohol-resistant foam, dry chemical or carbon dioxide.

Extinguishing media which must not be used for safety reasons

No information available.

Specific Hazards Arising from the Chemical

Thermal decomposition can lead to release of irritating gases and vapors. In the event of fire and/or explosion do not breathe fumes. Contact with metals may evolve flammable gas.

Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**Flash Point
Method**

Not applicable
No information available.

Autoignition Temperature

No information available.

SAF FIXATIVE

Revision Date 22-Sep-2009

6. ACCIDENTAL RELEASE MEASURES

Personal Precautions	Ensure adequate ventilation. Use personal protective equipment. Avoid contact with skin, eyes and clothing.
Environmental Precautions	Should not be released into the environment. Prevent further leakage or spillage if safe to do so. Do not flush into surface water or sanitary sewer system.
Methods for Containment and Clean Up	Soak up with inert absorbent material Keep in suitable and closed containers for disposal

7. HANDLING AND STORAGE

Handling	Wear personal protective equipment. Ensure adequate ventilation. Avoid contact with skin, eyes and clothing. Avoid ingestion and inhalation. Avoid contact with skin. Avoid contact with skin and eyes. Do not breathe dust. Pay attention to flashback. Take necessary action to avoid static electricity discharge (which might cause ignition of organic vapors).
Storage	Keep containers tightly closed in a dry, cool and well-ventilated place. Keep container tightly closed in a dry and well-ventilated place. Keep out of the reach of children. Keep containers tightly closed in a cool, well-ventilated place.
Specific use(s)	No information available.

8. EXPOSURE CONTROLS / PERSONAL PROTECTION

Exposure limits

Component	European Union	The United Kingdom	France	Spain	Germany
Formaldehyde			VME: 0.5 ppm VLCT: 1 ppm	VLA-EC: 0.37 mg/m ³ VLA-EC: 0.3 ppm	
Methyl alcohol	TWA: 200 ppm TWA: 260 mg/m ³ Skin		VME: 200 ppm VME: 260 mg/m ³ VLCT: 1000 ppm VLCT: 1300 mg/m ³	VLA-ED: 266 mg/m ³ VLA-ED: 200 ppm	
Acetic acid			VLCT: 10 ppm VLCT: 25 mg/m ³	VLA-EC: 37 mg/m ³ VLA-EC: 15 ppm VLA-ED: 10 ppm VLA-ED: 25 mg/m ³	MAK: 10 ppm MAK: 25 mg/m ³ Peak: 20 ppm Peak: 50 mg/m ³

Component	Italy	Portugal	The Netherlands	Finland	Austria
Formaldehyde		Ceiling: 0.3 ppm	STEL: 0.5 mg/m ³ TWA: 0.15 mg/m ³	TWA: 0.37 mg/m ³ TWA: 0.3 ppm STEL: 1.2 mg/m ³ STEL: 1 ppm Ceiling: 1 ppm Ceiling: 1.2 mg/m ³	STEL: 0.5 ppm STEL: 0.6 mg/m ³ MAK: 0.5 ppm MAK: 0.6 mg/m ³ Ceiling: 0.5 ppm Ceiling: 0.6 mg/m ³

SAF FIXATIVE
Revision Date 22-Sep-2009

Component	Italy	Portugal	The Netherlands	Finland	Austria
Methyl alcohol	TWA: 200 ppm TWA: 260 mg/m ³	STEL: 250 ppm TWA: 200 ppm	STEL: 520 mg/m ³ TWA: 260 mg/m ³	TWA: 270 mg/m ³ TWA: 200 ppm STEL: 330 mg/m ³ STEL: 250 ppm	STEL: 1040 mg/m ³ STEL: 800 ppm MAK: 260 mg/m ³ MAK: 200 ppm
Acetic acid		STEL: 15 ppm TWA: 10 ppm		TWA: 5 ppm TWA: 13 mg/m ³ STEL: 10 ppm STEL: 25 mg/m ³	STEL: 50 mg/m ³ STEL: 20 ppm MAK: 25 mg/m ³ MAK: 10 ppm

Component	Switzerland	Poland	Norway	Ireland	Denmark	Australia
Formaldehyde	STEL: 0.6 ppm STEL: 0.74 mg/m ³ MAK: 0.3 ppm MAK: 0.37 mg/m ³	NDSch: 1 mg/m ³ NDS: 0.5 mg/m ³	TWA: 0.5 ppm TWA: 0.6 mg/m ³ Ceiling: 1 ppm Ceiling: 1.2 mg/m ³	TWA: 2.5 mg/m ³ TWA: 2 ppm STEL: 2.5 mg/m ³ STEL: 2 ppm	Ceiling: 0.3 ppm Ceiling: 0.4 mg/m ³	STEL: 2 ppm STEL: 2.5 mg/m ³ TWA: 1.2 mg/m ³ TWA: 1 ppm
Methyl alcohol	STEL: 1040 mg/m ³ STEL: 800 ppm MAK: 200 ppm MAK: 260 mg/m ³	NDSch: 300 mg/m ³ NDS: 100 mg/m ³	TWA: 100 ppm TWA: 130 mg/m ³	TWA: 200 ppm TWA: 260 mg/m ³ STEL: 250 ppm STEL: 310 mg/m ³ Skin	TWA: 200 ppm TWA: 260 mg/m ³	STEL: 250 ppm STEL: 328 mg/m ³ TWA: 262 mg/m ³ TWA: 200 ppm
Acetic acid	STEL: 50 mg/m ³ STEL: 20 ppm MAK: 25 mg/m ³ MAK: 10 ppm	NDSch: 30 mg/m ³ NDS: 15 mg/m ³	TWA: 10 ppm TWA: 25 mg/m ³	TWA: 25 mg/m ³ TWA: 10 ppm STEL: 15 ppm STEL: 37 mg/m ³	TWA: 10 ppm TWA: 25 mg/m ³	STEL: 15 ppm STEL: 37 mg/m ³ TWA: 10 ppm TWA: 25 mg/m ³

Occupational exposure controls
Engineering Measures

Ensure that eyewash stations and safety showers are close to the workstation location Ensure adequate ventilation, especially in confined areas

Personal Protective Equipment
Respiratory Protection

Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

Eye Protection

Tightly fitting safety goggles

Skin and body protection

Wear appropriate protective gloves and clothing to prevent skin exposure. Lightweight protective clothing. Apron. Impervious gloves.

Hand Protection

Protective gloves

Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

Environmental exposure controls

No information available.

9. PHYSICAL AND CHEMICAL PROPERTIES
Physical State

Liquid

pH

No information available.

Boiling Point/Range

Not applicable

Melting Point/Range

No information available.

Flash Point

Not applicable

SAF FIXATIVE

Revision Date 22-Sep-2009

10. STABILITY AND REACTIVITY

Stability	Stable under normal conditions.
Conditions to Avoid	Incompatible products. Excess heat.
Incompatible Materials	Strong oxidizing agents
Hazardous Decomposition Products	Carbon monoxide (CO), Carbon dioxide (CO ₂)
Hazardous Polymerization	Hazardous polymerization does not occur.
Hazardous Reactions .	None under normal processing..

11. TOXICOLOGICAL INFORMATION

Acute Toxicity

Component Information

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Formaldehyde	500 mg/kg (Rat)		0.578 mg/L (Rat) 4 h
Methyl alcohol	5628 mg/kg (Rat)	15800 mg/kg (Rabbit)	64000 ppm (Rat) 4 h 83.2 mg/L (Rat) 4 h
Acetic acid	3310 mg/kg (Rat)	1060 mg/kg (Rabbit)	11.4 mg/L (Rat) 4 h
Sodium acetate	3530 mg/kg (Rat)	10 g/kg (Rabbit)	30 g/m ³ (Rat) 1 h

Chronic Toxicity

Carcinogenicity The table below indicates whether each agency has listed any ingredient as a carcinogen

Component	IARC	UK
Formaldehyde	Group 1	

IARC: (International Agency for Research on Cancer)

IARC: (International Agency for Research on Cancer)

Group 1 - Carcinogenic to Humans

Group 2A - Probably Carcinogenic to Humans

Group 2B - Possibly Carcinogenic to Humans

Sensitization	No information available.
Neurological Effects	No information available.
Mutagenic Effects	No information available.
Reproductive Effects	No information available.

SAF FIXATIVE
Revision Date 22-Sep-2009
Developmental Effects No information available.

Target Organs Gastrointestinal tract (GI), Central nervous system (CNS), Eyes, Respiratory system, Skin, Optic nerve, None known., Teeth.

12. ECOLOGICAL INFORMATION
Ecotoxicity

Component	Freshwater Algae	Freshwater Fish	Microtox	Water Flea
Formaldehyde				EC50 96 h 20 mg/L EC50 48 h 2 mg/L EC50 96 h 20 mg/L
Methyl alcohol			EC50 = 39000 mg/L 25 min EC50 = 40000 mg/L 15 min EC50 = 43000 mg/L 5 min	
Acetic acid			= 8.8 mg/L EC50 Photobacterium phosphoreum 15 min = 8.8 mg/L EC50 Photobacterium phosphoreum 25 min = 8.8 mg/L EC50 Photobacterium phosphoreum 5 min = 7200 mg/L EC50 Pseudomonas putida 18 h	EC50 24 h 95 mg/L
Sodium acetate				EC50 48 h 5800 mg/L

Persistence and Degradability No information available

Bioaccumulative Potential No information available.

Mobility No information available.

Component	log Pow
Formaldehyde	0.35
Methyl alcohol	-0.77
Acetic acid	-0.31

13. DISPOSAL CONSIDERATIONS
Waste from Residues / Unused Products Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification

Contaminated Packaging Dispose of in accordance with local regulations.

SAF FIXATIVE

Revision Date 22-Sep-2009

Other Information

According to the European Waste Catalogue, Waste Codes are not product specific, but application specific. Waste codes should be assigned by the user based on the application for which the product was used.

14. TRANSPORT INFORMATION

IMDG/IMO

Not regulated

ADR

Not regulated

IATA

Not regulated

15. REGULATORY INFORMATION

Classification according to European directive on classification of hazardous preparations 1999/45/EC

Labelling

Xn



R -phrase(s)

none

R40 - Limited evidence of a carcinogenic effect

R43 - May cause sensitization by skin contact

S -phrase(s)

S24/25 - Avoid contact with skin and eyes

International Inventories

Component	EINECS	ELINCS	NLP	TSCA	DSL	NDSL	PICCS	ENCS	CHINA	AICS	KECL
Formaldehyde	200-001-8	-		X	X	-	X	X	X	X	KE-17074 X
Methyl alcohol	200-659-6	-		X	X	-	X	X	X	X	KE-23193 X
Acetic acid	200-580-7	-		X	X	-	X	X	X	X	KE-00013 X
Sodium acetate	204-823-8	-		X	X	-	X	X	X	X	KE-00061 X

SAF FIXATIVE

Revision Date 22-Sep-2009

16. OTHER INFORMATION

Text of R phrases mentioned in Section 2-3

R10 - Flammable

R11 - Highly flammable

R34 - Causes burns

R35 - Causes severe burns

R40 - Limited evidence of a carcinogenic effect

R43 - May cause sensitization by skin contact

R23/24/25 - Toxic by inhalation, in contact with skin and if swallowed

R39/23/24/25 - Toxic: danger of very serious irreversible effects through inhalation, in contact with skin and if swallowed

Prepared By	Regulatory Affairs
Creation Date	22-Sep-2009
Revision Date	22-Sep-2009
Revision Summary	No information available..
Restrictions on use	No information available.
Training advice	No information available.
Literary reference	No information available.

Disclaimer

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End of Safety Data Sheet